

Parameter	Description	Type	Size	Default
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0
start-ip	Start of IP range.	ipv6-address	Not Specified	::

config prefix-range

Parameter	Description	Type	Size	Default
end-prefix	End of prefix range.	ipv6-address	Not Specified	::
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0
prefix-length	Prefix length.	integer	Minimum value: 1 Maximum value: 128	0
start-prefix	Start of prefix range.	ipv6-address	Not Specified	::

config system dhcp server

Configure DHCP servers.

```
config system dhcp server
  Description: Configure DHCP servers.
  edit <id>
    set auto-configuration [disable|enable]
    set auto-managed-status [disable|enable]
    set conflicted-ip-timeout {integer}
    set ddns-auth [disable|tsig]
    set ddns-key {password_aes256}
    set ddns-keyname {string}
    set ddns-server-ip {ipv4-address}
    set ddns-ttl {integer}
    set ddns-update [disable|enable]
    set ddns-update-override [disable|enable]
    set ddns-zone {string}
    set default-gateway {ipv4-address}
    set dhcp-settings-from-fortiipam [disable|enable]
    set dns-server1 {ipv4-address}
```

```

set dns-server2 {ipv4-address}
set dns-server3 {ipv4-address}
set dns-server4 {ipv4-address}
set dns-service [local|default|...]
set domain {string}
config exclude-range
    Description: Exclude one or more ranges of IP addresses from being assigned to
clients.
    edit <id>
        set end-ip {ipv4-address}
        set lease-time {integer}
        set start-ip {ipv4-address}
        set uci-match [disable|enable]
        set uci-string <uci-string1>, <uci-string2>, ...
        set vci-match [disable|enable]
        set vci-string <vci-string1>, <vci-string2>, ...
    next
end
set filename {string}
set forticlient-on-net-status [disable|enable]
set interface {string}
set ip-mode [range|usrgrp]
config ip-range
    Description: DHCP IP range configuration.
    edit <id>
        set end-ip {ipv4-address}
        set lease-time {integer}
        set start-ip {ipv4-address}
        set uci-match [disable|enable]
        set uci-string <uci-string1>, <uci-string2>, ...
        set vci-match [disable|enable]
        set vci-string <vci-string1>, <vci-string2>, ...
    next
end
set ipsec-lease-hold {integer}
set lease-time {integer}
set mac-acl-default-action [assign|block]
set netmask {ipv4-netmask}
set next-server {ipv4-address}
set ntp-server1 {ipv4-address}
set ntp-server2 {ipv4-address}
set ntp-server3 {ipv4-address}
set ntp-service [local|default|...]
config options
    Description: DHCP options.
    edit <id>
        set code {integer}
        set ip {user}
        set type [hex|string|...]
        set uci-match [disable|enable]
        set uci-string <uci-string1>, <uci-string2>, ...
        set value {string}
        set vci-match [disable|enable]
        set vci-string <vci-string1>, <vci-string2>, ...
    next
end

```

```

set relay-agent {ipv4-address}
config reserved-address
    Description: Options for the DHCP server to assign IP settings to specific MAC
addresses.
    edit <id>
        set action [assign|block|...]
        set circuit-id {string}
        set circuit-id-type [hex|string]
        set description {var-string}
        set ip {ipv4-address}
        set mac {mac-address}
        set remote-id {string}
        set remote-id-type [hex|string]
        set type [mac|option82]
    next
end
set server-type [regular|ipsec]
set shared-subnet [disable|enable]
set status [disable|enable]
set tftp-server <tftp-server1>, <tftp-server2>, ...
set timezone {string}
set timezone-option [disable|default|...]
set vci-match [disable|enable]
set vci-string <vci-string1>, <vci-string2>, ...
set wifi-ac-service [specify|local]
set wifi-ac1 {ipv4-address}
set wifi-ac2 {ipv4-address}
set wifi-ac3 {ipv4-address}
set wins-server1 {ipv4-address}
set wins-server2 {ipv4-address}
next
end

```

config system dhcp server

Parameter	Description	Type	Size	Default
auto-configuration	Enable/disable auto configuration.	option	-	enable
	Option	Description		
	<i>disable</i>	Disable auto configuration.		
	<i>enable</i>	Enable auto configuration.		
auto-managed-status	Enable/disable use of this DHCP server once this interface has been assigned an IP address from FortiIPAM.	option	-	enable

Parameter	Description	Type	Size	Default
	Option	Description		
	<i>disable</i>	Disable use of this DHCP server once this interface has been assigned an IP address from FortiIPAM.		
	<i>enable</i>	Enable use of this DHCP server once this interface has been assigned an IP address from FortiIPAM.		
conflicted-ip-timeout	Time in seconds to wait after a conflicted IP address is removed from the DHCP range before it can be reused.	integer	Minimum value: 60 Maximum value: 8640000	1800
ddns-auth	DDNS authentication mode.	option	-	disable
	Option	Description		
	<i>disable</i>	Disable DDNS authentication.		
	<i>tsig</i>	TSIG based on RFC2845.		
ddns-key	DDNS update key (base 64 encoding).	password_aes256	Not Specified	
ddns-keyname	DDNS update key name.	string	Maximum length: 64	
ddns-server-ip	DDNS server IP.	ipv4-address	Not Specified	0.0.0.0
ddns-ttl	TTL.	integer	Minimum value: 60 Maximum value: 86400	300
ddns-update	Enable/disable DDNS update for DHCP.	option	-	disable
	Option	Description		
	<i>disable</i>	Disable DDNS update for DHCP.		
	<i>enable</i>	Enable DDNS update for DHCP.		
ddns-update-override	Enable/disable DDNS update override for DHCP.	option	-	disable
	Option	Description		
	<i>disable</i>	Disable DDNS update override for DHCP.		
	<i>enable</i>	Enable DDNS update override for DHCP.		

Parameter	Description	Type	Size	Default								
ddns-zone	Zone of your domain name (ex. DDNS.com).	string	Maximum length: 64									
default-gateway	Default gateway IP address assigned by the DHCP server.	ipv4-address	Not Specified	0.0.0.0								
dhcp-settings-from-fortiipam	Enable/disable populating of DHCP server settings from FortiIPAM.	option	-	disable								
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable populating of DHCP server settings from FortiIPAM.</td></tr><tr><td>enable</td><td>Enable populating of DHCP server settings from FortiIPAM.</td></tr></table>				Option	Description	disable	Disable populating of DHCP server settings from FortiIPAM.	enable	Enable populating of DHCP server settings from FortiIPAM.		
Option	Description											
disable	Disable populating of DHCP server settings from FortiIPAM.											
enable	Enable populating of DHCP server settings from FortiIPAM.											
dns-server1	DNS server 1.	ipv4-address	Not Specified	0.0.0.0								
dns-server2	DNS server 2.	ipv4-address	Not Specified	0.0.0.0								
dns-server3	DNS server 3.	ipv4-address	Not Specified	0.0.0.0								
dns-server4	DNS server 4.	ipv4-address	Not Specified	0.0.0.0								
dns-service	Options for assigning DNS servers to DHCP clients.	option	-	specify								
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>local</td><td>IP address of the interface the DHCP server is added to becomes the client's DNS server IP address.</td></tr><tr><td>default</td><td>Clients are assigned the FortiGate's configured DNS servers.</td></tr><tr><td>specify</td><td>Specify up to 3 DNS servers in the DHCP server configuration.</td></tr></table>				Option	Description	local	IP address of the interface the DHCP server is added to becomes the client's DNS server IP address.	default	Clients are assigned the FortiGate's configured DNS servers.	specify	Specify up to 3 DNS servers in the DHCP server configuration.
Option	Description											
local	IP address of the interface the DHCP server is added to becomes the client's DNS server IP address.											
default	Clients are assigned the FortiGate's configured DNS servers.											
specify	Specify up to 3 DNS servers in the DHCP server configuration.											
domain	Domain name suffix for the IP addresses that the DHCP server assigns to clients.	string	Maximum length: 35									
filename	Name of the boot file on the TFTP server.	string	Maximum length: 127									
forticlient-on-net-status	Enable/disable FortiClient-On-Net service for this DHCP server.	option	-	enable								
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable FortiClient On-Net Status.</td></tr><tr><td>enable</td><td>Enable FortiClient On-Net Status.</td></tr></table>				Option	Description	disable	Disable FortiClient On-Net Status.	enable	Enable FortiClient On-Net Status.		
Option	Description											
disable	Disable FortiClient On-Net Status.											
enable	Enable FortiClient On-Net Status.											
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0								

Parameter	Description	Type	Size	Default						
interface	DHCP server can assign IP configurations to clients connected to this interface.	string	Maximum length: 15							
ip-mode	Method used to assign client IP.	option	-	range						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>range</td><td>Use range defined by start-ip/end-ip to assign client IP.</td></tr><tr><td>usrgrp</td><td>Use user-group defined method to assign client IP.</td></tr></table>				Option	Description	range	Use range defined by start-ip/end-ip to assign client IP.	usrgrp	Use user-group defined method to assign client IP.
	Option	Description								
	range	Use range defined by start-ip/end-ip to assign client IP.								
usrgrp	Use user-group defined method to assign client IP.									
ipsec-lease-hold	DHCP over IPsec leases expire this many seconds after tunnel down (0 to disable forced-expiry).	integer	Minimum value: 0 Maximum value: 8640000	60						
lease-time	Lease time in seconds, 0 means unlimited.	integer	Minimum value: 300 Maximum value: 8640000	604800						
mac-acl-default-action	MAC access control default action (allow or block assigning IP settings).	option	-	assign						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>assign</td><td>Allow the DHCP server to assign IP settings to clients on the MAC access control list.</td></tr><tr><td>block</td><td>Block the DHCP server from assigning IP settings to clients on the MAC access control list.</td></tr></table>				Option	Description	assign	Allow the DHCP server to assign IP settings to clients on the MAC access control list.	block	Block the DHCP server from assigning IP settings to clients on the MAC access control list.
	Option	Description								
	assign	Allow the DHCP server to assign IP settings to clients on the MAC access control list.								
block	Block the DHCP server from assigning IP settings to clients on the MAC access control list.									
netmask	Netmask assigned by the DHCP server.	ipv4-netmask	Not Specified	0.0.0.0						
next-server	IP address of a server (for example, a TFTP sever) that DHCP clients can download a boot file from.	ipv4-address	Not Specified	0.0.0.0						
ntp-server1	NTP server 1.	ipv4-address	Not Specified	0.0.0.0						
ntp-server2	NTP server 2.	ipv4-address	Not Specified	0.0.0.0						
ntp-server3	NTP server 3.	ipv4-address	Not Specified	0.0.0.0						
ntp-service	Options for assigning Network Time Protocol (NTP) servers to DHCP clients.	option	-	specify						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>local</td><td>IP address of the interface the DHCP server is added to becomes the client's NTP server IP address.</td></tr></table>				Option	Description	local	IP address of the interface the DHCP server is added to becomes the client's NTP server IP address.		
	Option	Description								
local	IP address of the interface the DHCP server is added to becomes the client's NTP server IP address.									

Parameter	Description	Type	Size	Default
	Option	Description		
	<i>default</i>	Clients are assigned the FortiGate's configured NTP servers.		
	<i>specify</i>	Specify up to 3 NTP servers in the DHCP server configuration.		
relay-agent	Relay agent IP.	ipv4-address	Not Specified	0.0.0.0
server-type	DHCP server can be a normal DHCP server or an IPsec DHCP server.	option	-	regular
	Option	Description		
	<i>regular</i>	Regular DHCP service.		
	<i>ipsec</i>	DHCP over IPsec service.		
shared-subnet	Enable/disable shared subnet.	option	-	disable
	Option	Description		
	<i>disable</i>	Disable shared subnet.		
	<i>enable</i>	Enable shared subnet.		
status	Enable/disable this DHCP configuration.	option	-	enable
	Option	Description		
	<i>disable</i>	Do not use this DHCP server configuration.		
	<i>enable</i>	Use this DHCP server configuration.		
tftp-server <tftp-server>	One or more hostnames or IP addresses of the TFTP servers in quotes separated by spaces. TFTP server.	string	Maximum length: 63	
timezone	Select the time zone to be assigned to DHCP clients.	string	Maximum length: 63	
timezone-option	Options for the DHCP server to set the client's time zone.	option	-	disable
	Option	Description		
	<i>disable</i>	Do not set the client's time zone.		
	<i>default</i>	Clients are assigned the FortiGate's configured time zone.		
	<i>specify</i>	Specify the time zone to be assigned to DHCP clients.		

Parameter	Description	Type	Size	Default						
vci-match	Enable/disable vendor class identifier (VCI) matching. When enabled only DHCP requests with a matching VCI are served.	option	-	disable						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td><i>disable</i></td><td>Disable VCI matching.</td></tr><tr><td><i>enable</i></td><td>Enable VCI matching.</td></tr></table>	Option	Description	<i>disable</i>	Disable VCI matching.	<i>enable</i>	Enable VCI matching.			
	Option	Description								
	<i>disable</i>	Disable VCI matching.								
<i>enable</i>	Enable VCI matching.									
vci-string <vci-string>	One or more VCI strings in quotes separated by spaces. VCI strings.	string	Maximum length: 255							
wifi-ac-service	Options for assigning WiFi access controllers to DHCP clients.	option	-	specify						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td><i>specify</i></td><td>Specify up to 3 WiFi Access Controllers in the DHCP server configuration.</td></tr><tr><td><i>local</i></td><td>IP address of the interface the DHCP server is added to becomes the client's WiFi Access Controller IP address.</td></tr></table>	Option	Description	<i>specify</i>	Specify up to 3 WiFi Access Controllers in the DHCP server configuration.	<i>local</i>	IP address of the interface the DHCP server is added to becomes the client's WiFi Access Controller IP address.			
	Option	Description								
	<i>specify</i>	Specify up to 3 WiFi Access Controllers in the DHCP server configuration.								
<i>local</i>	IP address of the interface the DHCP server is added to becomes the client's WiFi Access Controller IP address.									
wifi-ac1	WiFi Access Controller 1 IP address (DHCP option 138, RFC 5417).	ipv4-address	Not Specified	0.0.0.0						
wifi-ac2	WiFi Access Controller 2 IP address (DHCP option 138, RFC 5417).	ipv4-address	Not Specified	0.0.0.0						
wifi-ac3	WiFi Access Controller 3 IP address (DHCP option 138, RFC 5417).	ipv4-address	Not Specified	0.0.0.0						
wins-server1	WINS server 1.	ipv4-address	Not Specified	0.0.0.0						
wins-server2	WINS server 2.	ipv4-address	Not Specified	0.0.0.0						

config exclude-range

Parameter	Description	Type	Size	Default
end-ip	End of IP range.	ipv4-address	Not Specified	0.0.0.0
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0

Parameter	Description	Type	Size	Default						
lease-time	Lease time in seconds, 0 means default lease time.	integer	Minimum value: 300 Maximum value: 8640000	0						
start-ip	Start of IP range.	ipv4-address	Not Specified	0.0.0.0						
uci-match	Enable/disable user class identifier (UCI) matching. When enabled only DHCP requests with a matching UCI are served with this range.	option	-	disable						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable UCI matching.</td></tr><tr><td>enable</td><td>Enable UCI matching.</td></tr></table>				Option	Description	disable	Disable UCI matching.	enable	Enable UCI matching.
Option	Description									
disable	Disable UCI matching.									
enable	Enable UCI matching.									
uci-string <uci-string>	One or more UCI strings in quotes separated by spaces. UCI strings.	string	Maximum length: 255							
vci-match	Enable/disable vendor class identifier (VCI) matching. When enabled only DHCP requests with a matching VCI are served with this range.	option	-	disable						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable VCI matching.</td></tr><tr><td>enable</td><td>Enable VCI matching.</td></tr></table>				Option	Description	disable	Disable VCI matching.	enable	Enable VCI matching.
Option	Description									
disable	Disable VCI matching.									
enable	Enable VCI matching.									
vci-string <vci-string>	One or more VCI strings in quotes separated by spaces. VCI strings.	string	Maximum length: 255							

config ip-range

Parameter	Description	Type	Size	Default
end-ip	End of IP range.	ipv4-address	Not Specified	0.0.0.0
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0

Parameter	Description	Type	Size	Default						
lease-time	Lease time in seconds, 0 means default lease time.	integer	Minimum value: 300 Maximum value: 8640000	0						
start-ip	Start of IP range.	ipv4-address	Not Specified	0.0.0.0						
uci-match	Enable/disable user class identifier (UCI) matching. When enabled only DHCP requests with a matching UCI are served with this range.	option	-	disable						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable UCI matching.</td></tr><tr><td>enable</td><td>Enable UCI matching.</td></tr></table>				Option	Description	disable	Disable UCI matching.	enable	Enable UCI matching.
Option	Description									
disable	Disable UCI matching.									
enable	Enable UCI matching.									
uci-string <uci-string>	One or more UCI strings in quotes separated by spaces. UCI strings.	string	Maximum length: 255							
vci-match	Enable/disable vendor class identifier (VCI) matching. When enabled only DHCP requests with a matching VCI are served with this range.	option	-	disable						
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable VCI matching.</td></tr><tr><td>enable</td><td>Enable VCI matching.</td></tr></table>				Option	Description	disable	Disable VCI matching.	enable	Enable VCI matching.
Option	Description									
disable	Disable VCI matching.									
enable	Enable VCI matching.									
vci-string <vci-string>	One or more VCI strings in quotes separated by spaces. VCI strings.	string	Maximum length: 255							

config options

Parameter	Description	Type	Size	Default
code	DHCP option code.	integer	Minimum value: 0 Maximum value: 255	0
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0

Parameter	Description	Type	Size	Default										
ip	DHCP option IPs.	user	Not Specified											
type	DHCP option type.	option	-	hex										
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>hex</td><td>DHCP option in hex.</td></tr><tr><td>string</td><td>DHCP option in string.</td></tr><tr><td>ip</td><td>DHCP option in IP.</td></tr><tr><td>fqdn</td><td>DHCP option in domain search option format.</td></tr></table>	Option	Description	hex	DHCP option in hex.	string	DHCP option in string.	ip	DHCP option in IP.	fqdn	DHCP option in domain search option format.			
Option	Description													
hex	DHCP option in hex.													
string	DHCP option in string.													
ip	DHCP option in IP.													
fqdn	DHCP option in domain search option format.													
uci-match	Enable/disable user class identifier (UCI) matching. When enabled only DHCP requests with a matching UCI are served with this option.	option	-	disable										
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable UCI matching.</td></tr><tr><td>enable</td><td>Enable UCI matching.</td></tr></table>	Option	Description	disable	Disable UCI matching.	enable	Enable UCI matching.							
Option	Description													
disable	Disable UCI matching.													
enable	Enable UCI matching.													
uci-string <uci-string>	One or more UCI strings in quotes separated by spaces. UCI strings.	string	Maximum length: 255											
value	DHCP option value.	string	Maximum length: 312											
vci-match	Enable/disable vendor class identifier (VCI) matching. When enabled only DHCP requests with a matching VCI are served with this option.	option	-	disable										
	<table><tr><th>Option</th><th>Description</th></tr><tr><td>disable</td><td>Disable VCI matching.</td></tr><tr><td>enable</td><td>Enable VCI matching.</td></tr></table>	Option	Description	disable	Disable VCI matching.	enable	Enable VCI matching.							
Option	Description													
disable	Disable VCI matching.													
enable	Enable VCI matching.													
vci-string <vci-string>	One or more VCI strings in quotes separated by spaces. VCI strings.	string	Maximum length: 255											

config reserved-address

Parameter	Description	Type	Size	Default
action	Options for the DHCP server to configure the client with the reserved MAC address.	option	-	reserved

Parameter	Description	Type	Size	Default
	Option	Description		
	<i>assign</i>	Configure the client with this MAC address like any other client.		
	<i>block</i>	Block the DHCP server from assigning IP settings to the client with this MAC address.		
	<i>reserved</i>	Assign the reserved IP address to the client with this MAC address.		
circuit-id	Option 82 circuit-ID of the client that will get the reserved IP address.	string	Maximum length: 312	
circuit-id-type	DHCP option type.	option	-	string
	Option	Description		
	<i>hex</i>	DHCP option in hex.		
	<i>string</i>	DHCP option in string.		
description	Description.	var-string	Maximum length: 255	
id	ID.	integer	Minimum value: 0 Maximum value: 4294967295	0
ip	IP address to be reserved for the MAC address.	ipv4-address	Not Specified	0.0.0.0
mac	MAC address of the client that will get the reserved IP address.	mac-address	Not Specified	00:00:00:00:00:00
remote-id	Option 82 remote-ID of the client that will get the reserved IP address.	string	Maximum length: 312	
remote-id-type	DHCP option type.	option	-	string
	Option	Description		
	<i>hex</i>	DHCP option in hex.		
	<i>string</i>	DHCP option in string.		
type	DHCP reserved-address type.	option	-	mac
	Option	Description		
	<i>mac</i>	Match with MAC address.		
	<i>option82</i>	Match with DHCP option 82.		